

James Westhead - Software Engineer

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GitHub: <https://github.com/WestheadJ>

Willing to work remotely or relocate within the UK

Personal Statement

BEng (Hons) Software Engineering graduate (2:1) with experience delivering full-stack web applications and data tools, with a strong focus on back-end development, debugging, and testing. Comfortable working in project teams and with real stakeholders, including a client-facing intelligence mapping tool built for Raytheon as part of a university project. Proven ability to balance shift leadership responsibilities in hospitality with full-time study, demonstrating resilience, ownership, and strong communication. Seeking junior/graduate software engineering roles, particularly in web development and back-end services.

Technical Skills

Fundamental Languages: Python, JavaScript C#, SQL

Familiar Programming Languages: PHP, Swift, Kotlin, Java

Frameworks & Libraries: [React.js](#), [Node.js](#), Flask, .NET, SvelteKit, Spectre Console, Pandas, React Router

Database: SQLite, MySQL, Supabase, Firebase, PocketBase

Tools: Git, VS Code, Visual Studio, Xcode, Netlify, Vercel, Expo

Practices: Unit testing, Agile, API integration, OOP, debuggin

Projects

Product Sales Tracker - Python CLI (Current)

- Developing a Python CLI tool using Spectre Console, Pandas, and DuckDB to analyse sales and trends at my current workplace..
- Using the Pandas library to clean and reshape complex report exports, then load them into DuckDB while preserving the relationships between tables.
- Enables fast querying and reporting on sales performance to support data-driven decisions at work

GitHub [Link](#)

Exception Analysis Tool - Final Year Dissertation (Jan-May 2025)

- Designed a C# tool to surface unhandled and potential runtime exceptions before deployment, aiming to reduce production errors.
- Explored .NET reflection and exception metadata to detect missed exceptions in code paths, including those thrown by library calls
- Faced limitations handling custom exceptions and third-party libraries, and struggled to distribute the tool reliably to users, meaning the implementation didn't fully validate the concept.

- Achieved a strong grade primarily due to the concept, design, and critical analysis of the approach, demonstrating deep technical exploration and honest evaluation of the prototype's limitations.

GitHub [Link](#)

Apple Music to Spotify Playlist Converter - Python

- Built a Python tool to convert Apple Music playlists into Spotify Playlists using web scraping and the Spotify Web API, all for free.
- Used requests and BeautifulSoup libraries to parse Apple Music playlist pages, and extract song/artist pairs without needing a paid Apple developer account.
- Integrated Spotify to authenticate with Spotify, create playlists, search for tracks, and add matches via the Spotify API.
- Implemented logging for unmatched tracks into timestamped text files, improving robustness and user feedback.

GitHub [Link](#)

Intelligence Mapping Tool - University Client Project for Raytheon

- Managed a team project to build a web application that calculated the percentage of an area occupied by Raytheon assets using a Raytheon test API.
- Implemented interactive maps and data visualisations with SvelteKit, MapLibre, [Chart.js](#) and [Turf.js](#), consuming REST APIs.
- Presented the solution to Raytheon engineers, demonstrating client-facing communication, teamwork, and the ability to deliver to external stakeholders.

GitHub [Link](#)

Relevant Experience

Koder.ly Web Development Work Experience

Oldham, UK (December 2019)

- Built a simple music-player website using HTML, CSS, and JavaScript during a two-week placement.
- Learned how to publish locally using IIS and accessed the site across devices on the same network.
- Observed a real engineering workflow including stand-ups, ticket systems (support + dev), and iterative task delivery.
- Gained early insight into agile communication, structured teamwork, and demoing completed work.

Raytheon Intelligence Mapping Tool - Designer / Developer / Project Manager

Sheffield Hallam - (2nd Year 1st Semester Module)

- Led a small team developing a geospatial intelligence tool using Svelte, MapLibre, Chart.js, and Turf.js.
- Communicated directly with Raytheon staff, gathering requirements and presenting sprint updates.
- Managed workload distribution, UI/UX design, feature prioritisation, and client-facing demonstrations.
- Strengthened skills in collaboration, data visualisation, problem-solving, and agile project delivery.

Tyre Ordering System – Project Manager

Sheffield Hallam - (3rd year 1st Semester Module)

- Designed a tyre ordering system for a consultancy, following PRINCE2-style project management.

- Produced wireframes, plans, and workflows using Figma and Microsoft Project.
- Coordinated the “client-style” project team, ensuring requirements were met and documentation remained aligned.

Work Experience

J D Wetherspoon - *Shift Leader / Team Leader/ Bar Associate*

Shaw, Sheffield, Oldham, UK (May 2023 – Present)

- Promoted through multiple roles in 2 years, demonstrating leadership, reliability, and ability to manage teams under pressure.
- Coordinated and led staff during shifts, ensuring smooth operations, excellent customer service, and timely problem resolution.
- Mentored and trained new staff, improving team efficiency and communication.
- Managed scheduling, conflict resolution, and operational decisions in a fast-paced environment.

The North Star Pub - Kitchen Porter

Shaw, UK (February 2021 - September 2022)

- Assisted with kitchen operations, demonstrating reliability, teamwork, and ability to work under pressure.

Education

Sheffield Hallam University — Sheffield, UK

[2:1] BEng (Hons) Software Engineering (*September 2022 – July 2025, Awarded July 2025*)

- Modules: Introduction to Software Engineering, Professional Software Projects, Development Project (Final year project, dissertation equivalent), Software Architecture and Design, Web Development, Databases and System Modelling, Systems Programming, Game Development, Management of IT Projects, Robotics

Ashton Sixth Form College - Ashton-under-Lyne, UK

A Level/BTEC Level 3 (*September 2020 - August 2022*)

- BTEC Level 3 Extended Diploma in Creative Media (Games Development) - Distinction*
- BTEC Level 3 Diploma in Music Technology - Distinction
- A-Level Computer Science - B

GCSEs - Oldham, UK

Achieved 6's and above including: English [6] | Maths [6] | Computer Science [6]